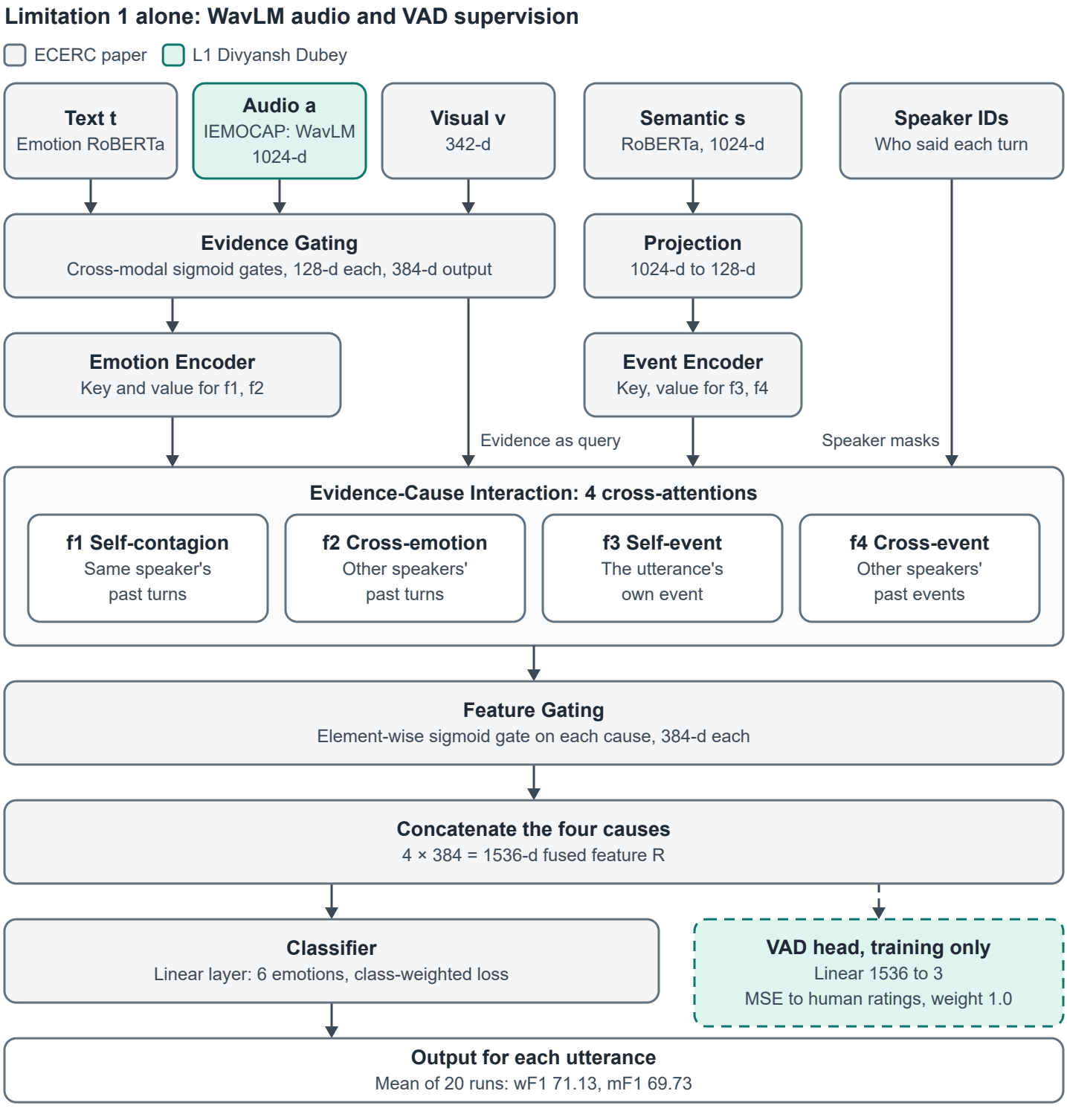

Limitation 1 alone: WavLM audio and VAD supervision
ECERC paper
L1 Divyansh Dubey
Text t
Emotion RoBERTa
Audio a
IEMOCAP: WavLM
1024-d
Visual v
342-d
Semantic s
RoBERTa, 1024-d
Speaker IDs
Who said each turn
Evidence Gating
Cross-modal sigmoid gates, 128-d each, 384-d output
Projection
1024-d to 128-d
Emotion Encoder
Key and value for f1, f2
Event Encoder
Key, value for f3, f4
Evidence as query
Speaker masks
Evidence-Cause Interaction: 4 cross-attentions
f1 Self-contagion
Same speaker's
past turns
f2 Cross-emotion
Other speakers'
past turns
f3 Self-event
The utterance's
own event
f4 Cross-event
Other speakers'
past events
Feature Gating
Element-wise sigmoid gate on each cause, 384-d each
Concatenate the four causes
4 × 384 = 1536-d fused feature R
Classifier
Linear layer: 6 emotions, class-weighted loss
VAD head, training only
Linear 1536 to 3
MSE to human ratings, weight 1.0
Output for each utterance
Mean of 20 runs: wF1 71.13, mF1 69.73